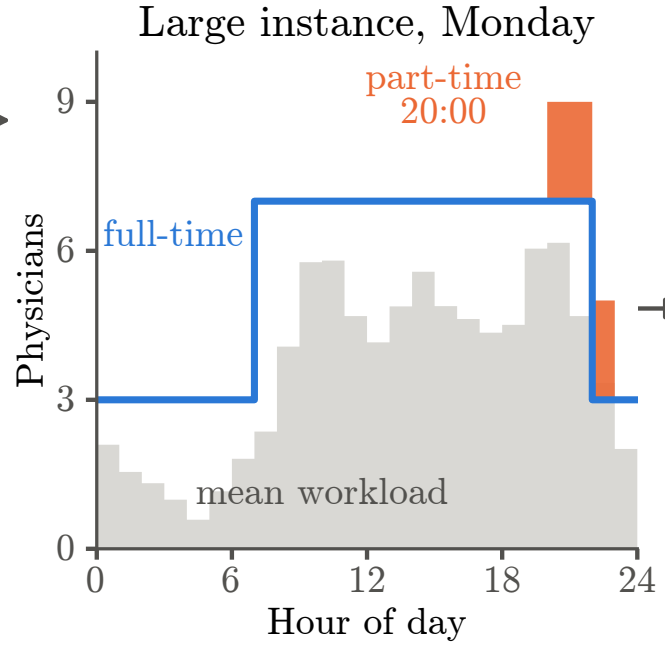
Phase 1: SO-HGASA

Simulated ED: every consultation needs a physician and a nurse → full- and part-time staff per shift

Phase 2: MILP + HGASA

→ weekly roster with the fewest staff



Optimal plan found in all ten enumerated cases

Part-time shifts cut waiting by 41% at full load

Optimal rosters in seconds; HGASA within 2.5% at 8× the size